

SGA 5 – Working Draft

Lefschetz formula and local terms; J -adic projective systems; beginning of ℓ -adic cohomology

Working English LaTeX translation

Exposé III B. Calculation of local terms – continuation

5.11. Twisted traces

Let A be a K -algebra, let $E \in \text{ob } D^-(A)$, and let $\rho : A \rightarrow A$ be an endomorphism of K -algebras. Denote by A_ρ the ring A , regarded as a left A^ρ -module through

$$a \otimes a' \longmapsto \rho(a)a'.$$

There is a canonical isomorphism

$$\text{Hom}_A(E, A_\rho \otimes_A E) \xrightarrow{\sim} \text{Hom}_A(E, E),$$

where the term on the right means the set of left A -module homomorphisms $E \rightarrow E$ which are ρ -linear, that is, such that

$$u(ax) = \rho(a)u(x).$$

For $u \in \text{Hom}_A(E, A_\rho \otimes_A E)$ we shall sometimes write $\text{Tr}_A^\rho(u)$, or simply $\text{Tr}_\rho(u)$, for the element $\text{Tr}_A(u) \in A_K$.

When ρ is an inner automorphism, say $\rho(a) = gag^{-1}$, one checks that $\text{Tr}_\rho(u)$ depends only on the class of g in A_K^* , and that for a ρ -linear u one has

$$\text{Tr}_\rho(u) = \text{the class of } g \text{ in } A_K \cdot \text{Tr}(g^{-1}u), \tag{5.13.1}$$

where $g^{-1}u$ is the A -linear endomorphism $x \mapsto g^{-1}u(x)$.

6. Equivariant correspondences and divisibility of local terms

6.1. Notation

We return to the notation of Part I and also fix a finite group G . We are given S -schemes with a left action of G , denoted $\underline{X}$ and $\underline{Y}$, together with stable closed subschemes

$$\underline{A}, \underline{B} \subset \underline{X} \times_S \underline{Y}.$$

We write X, Y, A, B for the underlying schemes, equivalently for the quotients by G in the sense used here. We assume that A and B satisfy the general-position hypotheses of I, 1.2. We are also given objects

$$L_X \in \text{ob } D(X), \quad L_Y \in \text{ob } D(Y),$$

carrying actions of G above the actions on X and Y , and G -equivariant homomorphisms

$$u : a_1^* L_X \longrightarrow a_2^! L_Y, \quad v : b_2^* L_Y \longrightarrow b_1^! L_X.$$

Then there are local terms

$$\langle u_z, v_z \rangle \in \Lambda$$

as in I, 1.1.1. On the other hand, the global cup-product $\langle \underline{u}, \underline{v} \rangle$ of I, 1.1.2 belongs to

$$H_c^4(X \times_S Y, K_{X \times_S Y}) = \Lambda,$$

and Theorem 1.2 applies to give

$$\langle u_z, v_z \rangle = \langle \underline{u}, \underline{v} \rangle.$$

One may also consider the non-commutative traces of u_z and v_z . More precisely, $(L_X)_x$ and $(L_Y)_y$ are perfect complexes of $\Lambda[G]$ -modules, and u_z, v_z are homomorphisms of $\Lambda[G]$ -modules. Thus, by 5.6.3, one can form

$$\text{Tr}_{\Lambda[G]}(u_z v_z) \in \Lambda[G]_{\Lambda},$$

where $\Lambda[G]_{\Lambda}$ is the free Λ -module on the set $G/\sim$ of conjugacy classes of G . Similarly the global cup-product gives an element

$$\langle\langle \underline{u}, \underline{v} \rangle\rangle \in H_c^4(X \times_S Y, \underline{\Lambda})_{\Lambda[G]},$$

where $\underline{\Lambda}$ denotes the constant sheaf with value $\Lambda[G]_{\Lambda}$.

Theorem 6.2. With the preceding notation, assume moreover that A and B are in general position in the sense of I, 1.2. Then

$$\mathrm{Tr}_{\Lambda[G]}(u_z v_z) = \langle\langle \underline{u}, \underline{v} \rangle\rangle.$$

In other words, the non-commutative local term coincides with the non-commutative global term.

The proof follows the same pattern as the proof of Theorem 1.2. By the same reductions as in no. 2, one is first reduced to the case where A and B are closed immersions; the matter is then to prove a vanishing analogous to (2.4.1). The key lemma is the following.

Lemma 6.2.1. Under the hypotheses of 6.2, suppose that $(L_X)|_U = 0$ and $(L_Y)|_V = 0$, with the notation of 2.1. Then

$$\mathrm{Tr}_{\Lambda[G]}(u_z v_z) = \mathrm{Tr}_{\Lambda[G]}(\underline{u} \cdot \underline{v})$$

in $\Lambda[G]_\Lambda$.

The proof of 6.2.1 is analogous to that of 2.3. One is reduced to the case in which L_X and L_Y are concentrated at the closed points, and one applies the non-commutative Lefschetz formula for the projections $\{x\} \rightarrow S$ and $\{y\} \rightarrow S$.

To finish the proof of 6.2 one still has to establish the non-commutative analogue of 2.5.

Proposition 6.2.2. Under the hypotheses of 6.2, suppose that a and b are closed immersions and that $(L_X)_x = 0$, $(L_Y)_y = 0$. Let T be the strict localization of $X \times_S Y$ at $z = (x, y)$, and identify A and B with closed subschemes of T . Then the restriction map

$$\mathrm{Tr}_{\Lambda[G]} : R\Gamma_A\left(T, \mathrm{RHom}(DL_X \otimes L_Y, K_T)\right) \longrightarrow R\Gamma\left(B, \mathrm{RHom}(DL_X \otimes L_Y, K_T)|_B\right)$$

is zero in $D(\Lambda[G]_\Lambda)$.

The proof occupies the rest of no. 6. As in no. 3, one is reduced to the tame case, where the representations of inertia factor through ℓ -groups. The reduction to the tame case is analogous to 3.1–3.3. The remaining tame case is treated by means of the purity theorem 4.3 and by an explicit calculation of local terms.

Theorem 6.2 has important consequences for the divisibility of local terms. Let χ be a virtual representation of G over Λ , i.e. an element of $R_\Lambda(G)$. One may consider the local term $\chi(u_z, v_z)$ obtained by applying χ to the non-commutative local term $\mathrm{Tr}_{\Lambda[G]}(u_z v_z)$. More

precisely, if $\chi = \sum n_i [V_i]$, with the V_i irreducible representations of G , put

$$\chi(u_z, v_z) = \sum n_i \operatorname{Tr}(u_z v_z \mid (L_X)_x \otimes_{\Lambda[G]} V_i).$$

Then 6.2 implies the following.

Corollary 6.3. With the notation of 6.2, let $\chi \in R_\Lambda(G)$, and write

$$e_\chi = \frac{1}{|G|} \sum_{g \in G} \chi(g^{-1}) g$$

for the central idempotent of $\Lambda[G]$ associated with χ . Then

$$\chi(u_z, v_z) = e_\chi \cdot \langle \underline{u}, \underline{v} \rangle.$$

In particular, if χ is a representation of degree 1, then

$$\chi(u_z, v_z) = \chi(\langle u_z, v_z \rangle^G),$$

where $\langle u_z, v_z \rangle^G$ denotes the component of $\operatorname{Tr}_{\Lambda[G]}(u_z v_z)$ on the trivial conjugacy class.

When G is cyclic, and one takes for L_X and L_Y the constant sheaves $\Lambda[G]$, and for u and v the correspondences given by left and right multiplication, Theorem 6.2 recovers Grothendieck's divisibility formula (XII, 4.5). More generally, for every finite group G and every virtual representation χ , formula 6.3 gives a canonical way of “dividing” ordinary local terms attached to equivariant correspondences, as announced in the introduction.

Complementary bibliography

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6.7. Traces and change of scalars

Let X and c be as in 6.5, and let $A \rightarrow B$ be a morphism of flat Λ -algebras. With respect to extension and restriction of scalars, the maps Tr_A of 6.5.3, 6.5.8, and 6.5.9 behave like the maps Tr_A of 5.6.1 and 5.6.2. The point is that, if f is an S -morphism, the functors

$$f^*, \quad f_*, \quad f!, \quad f^\dagger$$

commute with extension and restriction of scalars. Therefore the study of the behavior of Tr_A in 6.5.3 with respect to extension and restriction of scalars is reduced to the behavior of the tensor-product map

$$\mathrm{RHom}_A(p_1^{*L}, p_1^{*KA_X}) \otimes_{Ap_2^{*L}}^L \longrightarrow \mathrm{RHom}_A(p_1^{*L}, p_1^{*KA_X}) \otimes_{Ap_2^{*L}}^L$$

and of the evaluation map

$$\mathrm{RHom}_A(L, KA_X) \otimes_A^L L \longrightarrow KA_{X, L_{\mathfrak{h}}}.$$

Thus the commutativity of the squares 5.9.1 and 5.9.2 implies the commutativity of

$$\begin{array}{ccc} \delta^{*c} {}_*\mathrm{RHom}_A(c_1^{*L}, c_2^{!L}) & \xrightarrow{\mathrm{Tr}_A} & i_*^c KA_{X^c, L_{\mathfrak{h}}} \\ \downarrow & & \downarrow \\ \delta^{*c} {}_*\mathrm{RHom}_B(c_1^*(B \otimes_A^L L), c_2^!(B \otimes_A^L L)) & \xrightarrow{\mathrm{Tr}_B} & i_*^c KB_{X^c, L_{\mathfrak{h}}}. \end{array} \quad (6.7.1)$$

Here the left vertical map is extension of scalars and the right vertical map is defined by the canonical morphism

$$A \otimes_{A^e}^L A \longrightarrow B \otimes_{B^e}^L B.$$

It follows that, for $u \in \mathrm{Hom}_A(c_1^{*L}, c_2^{!L})$,

$$\mathrm{Tr}_B(B \otimes_A^L u) = \varphi \mathrm{Tr}_A(u), \quad (6.7.2)$$

where

$$\varphi : H^0(X^c, KA_{X^c, L_{\mathfrak{h}}}) \longrightarrow H^0(X^c, KB_{X^c, L_{\mathfrak{h}}})$$

is the canonical map.

Assume that A is of finite type and flat as a Λ -module, that B is of finite type and flat as a left A -module, and that $A_{\mathfrak{h}}$ and $B_{\mathfrak{h}}$ are flat over Λ . We then have $\mathrm{Tr}_{B/A} : B_{\mathfrak{h}} \rightarrow A_{\mathfrak{h}}$, and we again write

$$\mathrm{Tr}_{B/A} : KB_{X, \mathfrak{h}} \longrightarrow KA_{X, \mathfrak{h}} \quad (6.7.3)$$

for the map obtained by applying $K_X \overset{L}{\otimes} -$, as in 6.5.2. It follows from 5.10.6 and 5.10.9 that, for $L \in \text{ob } D_{ctf}^-(X)$, the square

$$\begin{array}{ccc} \delta^{*c} {}_*\text{RHom}_B(c_1^{*L}, c_2^{!L}) & \xrightarrow{\text{Tr}_B} & i_*^c K B_{X^c, \mathfrak{q}} \\ \downarrow & & \downarrow \text{Tr}_{B/A} \\ \delta^{*c} {}_*\text{RHom}_A(c_1^{*L}, c_2^{!L}) & \xrightarrow{\text{Tr}_A} & i_*^c K A_{X^c, \mathfrak{q}} \end{array} \quad (6.7.4)$$

is commutative, the left vertical map being restriction of scalars. Consequently, for $u \in \text{Hom}_B(c_1^{*L}, c_2^{!L})$, one has in $H^0(X^c, K A_{X^c, \mathfrak{q}})$

$$\text{Tr}_A(u) = \text{Tr}_{B/A}(\text{Tr}_B(u)). \quad (6.7.5)$$

6.8. External products

Let A and B be flat Λ -algebras, let X and Y be S -schemes, and put $Z = X \times_S Y$. Let

$$c : C \rightarrow X^2, \quad d : D \rightarrow Y^2$$

be correspondences, and let

$$e = c \times_S d : E = C \times_S D \rightarrow Z^2$$

be their product. Then $Z^e = X^c \times_S Y^d$. Put $R = A \otimes_\Lambda B$. As in III, 5.3, one defines the external tensor product

$$\text{Hom}_A(c_1^{*L}, c_2^{!L}) \otimes \text{Hom}_B(d_1^{*M}, d_2^{!M}) \xrightarrow{\overset{L}{\otimes}_S} \text{Hom}_R(e_1^*(L \overset{L}{\otimes}_S M), e_2^!(L \overset{L}{\otimes}_S M)). \quad (6.8.1)$$

From the commutativity of 5.12.1 and 5.12.3 it follows easily that, for $u \in \text{Hom}_A(c_1^{*L}, c_2^{!L})$ and $v \in \text{Hom}_B(d_1^{*M}, d_2^{!M})$, one has in $H^0(Z^e, K R_{Z^e, L_{\mathfrak{q}}})$

$$\text{Tr}_R(u \overset{L}{\otimes}_S v) = \text{Tr}_A(u) \text{Tr}_B(v), \quad (6.8.2)$$

where the product on the right is defined by the canonical map

$$H^0(X^c, K A_{X^c, L_{\mathfrak{q}}}) \otimes H^0(Y^d, K B_{Y^d, L_{\mathfrak{q}}}) \longrightarrow H^0(Z^e, K R_{Z^e, L_{\mathfrak{q}}}). \quad (6.8.3)$$

6.9. G -schemes and equivariant cohomological correspondences

Let G be a finite group. If X is a G - S -scheme, that is, an S -scheme endowed with an action of G by S -automorphisms, and if A is a Λ -algebra, we write $D(G, A^X)$ for the derived category of sheaves of left G - A -modules on X for the étale topology. For the notions of G -sheaves of sets, rings, modules, etc., we refer to XV, 1. We use the subscripts c and ctf for the full subcategories consisting of complexes which, as complexes of A -modules, have constructible cohomology, respectively finite Tor-dimension and constructible cohomology.

Similarly, if B is a Λ -algebra, we write $D(G, A_B^X)$ for the derived category of G -(A, B)-bimodules on X . If G acts trivially on X , then a left G - A -module on X is just a left $A[G]$ -module on X without operators. Thus $D(G, A^X) = D(A[G]^X)$. Since G is finite one also has $D_c(A^X) = D_c(A[G]^X)$, but only an inclusion

$$D_{ctf}(A[G]^X) \subset D_{ctf}(G, A^X).$$

The duality formalism of SGA 4, XVII–XVIII extends without difficulty to G - S -schemes. Since the functor forgetting the G -action is faithful, it is enough to extend naturally the definitions of the six operations

$${}^L\otimes, \quad \mathrm{RHom}, \quad f^*, \quad f_*, \quad f_!, \quad f^!$$

and of the canonical morphisms considered there (base change, Künneth, duality, etc.). This extension causes no difficulty; for the definition of $f_!$ and $f^!$ one uses equivariant compactifications. We leave to the reader the paraphrase of the definitions and results of 6.2–6.8 in the setting of G - S -schemes.

If, for $i = 1, 2$, X_i is a G - S -scheme, if $c : C \rightarrow X = X_1 \times_S X_2$ is a G - S -morphism, and if $L_i \in \mathrm{ob} D_{ctf}(G, A_{X_i}^{X_i})$, the elements of

$$\mathrm{Hom}_A(c_1^{*L_1}, c_2^{!L_2})$$

(where Hom_A means Hom in the relevant equivariant derived category) will be called *equivariant cohomological correspondences from L_1 to L_2 with support in c* . Notice that, when G acts trivially on the X_i and $L_i \in \mathrm{ob} D_{ctf}(A[G]^{X_i})$, the isomorphism 6.4.1 in $D(G, X)$ refines to an isomorphism

$$D_{A[G]}L_1 \stackrel{L}{\otimes}_{S, A[G]} L_2 \xrightarrow{\sim} \mathrm{RHom}_{A[G]}(p_1^{*L_1}, p_2^{!L_2}).$$

Consequently, in an equivariant situation of type 6.5, with G acting trivially on the spaces and with $L \in \mathrm{ob} D_{ctf}(A[G]^X) \subset D_{ctf}(G, A^X)$, one can, for $u \in \mathrm{Hom}_{A[G]}(c_1^{*L}, c_2^{!L})$, define a

local term

$$\mathrm{Tr}_{A[G]}(u) \in H^0(X^c, KA[G]_{X^c, L_{\mathfrak{h}}}),$$

finer in the evident sense than the local term

$$\mathrm{Tr}_A(u) \in H^0(G, X^c, KA_{X^c, L_{\mathfrak{h}}})$$

obtained by using only the finite Tor-dimension of L relative to A .

6.10. Values on conjugacy classes

Let X and c be as in 6.5, and let $L \in \mathrm{ob} D_{ctf}(A[G]^X)$. For $u \in \mathrm{Hom}_{A[G]}(c_1^{*L}, c_2^{!L})$, 6.5.9 gives a local trace

$$\mathrm{Tr}_{A[G]}(u) \in H^0(X^c, KA[G]_{X^c, \mathfrak{h}}) = H^0(X^c, K_{X^c} \otimes A[G_{\mathfrak{h}}]), \quad (6.10.1)$$

where $G_{\mathfrak{h}}$ is the set of conjugacy classes of G . For every $x \in G_{\mathfrak{h}}$ one may consider the value $\mathrm{Tr}_{A[G]}(u)(x) \in H^0(X^c, K_{X^c})$, obtained by projecting onto the component indexed by x . From 6.7.5 and 5.11.2 one obtains

$$\mathrm{Tr}_A(u) = |G| \mathrm{Tr}_{A[G]}(u)(e), \quad (6.10.2)$$

where e denotes the identity element of G .

More generally, let $g \in G$, let Z_g be the centralizer of g , and let $gu \in \mathrm{Hom}_{A[Z_g]}(c_1^{*L}, c_2^{!L})$ be the correspondence obtained by composing u with left multiplication by g on c_1^{*L} . Then $L \in \mathrm{ob} D_{ctf}(A[Z_g]^X)$, and 6.10.2 gives

$$\mathrm{Tr}_A(gu) = |Z_g| \mathrm{Tr}_{A[Z_g]}(gu)(e). \quad (6.10.3)$$

By the same argument as in 5.11.3, 6.7.4 gives

$$\mathrm{Tr}_{A[Z_g]}(gu)(e) = \mathrm{Tr}_{A[G]}(u)(g^{-1}), \quad (6.10.4)$$

and hence, if desired,

$$\mathrm{Tr}_A(gu) = |Z_g| \mathrm{Tr}_{A[G]}(u)(g^{-1}). \quad (6.10.5)$$

6.11–6.13. Direct images and ℓ -adic passage to the limit

Let $f : X \rightarrow Y$ be a proper G - S -morphism, and let

$$\begin{array}{ccc} X \times_S X & \xleftarrow{c} & C \\ \downarrow f \times_S f & & \downarrow \\ Y \times_S Y & \xleftarrow{d} & D \end{array} \quad (6.11.1)$$

be a commutative equivariant square of G - S -schemes, with c and d proper. Assume that G acts trivially on Y and on D . Write

$$f^{c/d}, \text{ or } f^c : X^c \longrightarrow Y^d, \quad (6.11.2)$$

for the morphism induced on schemes of fixed points. Let $L \in \text{ob } D_{ctf}(G, X)$ and let $u \in \text{Hom}(c_1^{*L}, c_2^{!L})$ be an equivariant correspondence with support in c . Its direct image is an equivariant correspondence

$$f_* u \in \text{Hom}_{A[G]}(d_1^* f_* L, d_2^! f_* L).$$

For $g \in G$, write Z_g for the centralizer of g , and let

$$gu \in \text{Hom}((gc)_1^{*L}, c_2^{!L}) \quad (6.11.3)$$

be the Z_g -equivariant correspondence with support in

$$gc = (gc_1 = c_1 g, c_2) : C \longrightarrow X \times_S X \quad (6.11.4)$$

defined as the composite

$$g^*(c_1^{*L}) \xrightarrow{\alpha_g} c_1^{*L} \xrightarrow{u} c_2^{!L},$$

where α_g is the map giving the action of G on c_1^{*L} . It is clear that

$$f_*(gu) = g(f_* u). \quad (6.11.5)$$

Proposition 6.12. Under the hypotheses of 6.11, assume that $f_* L \in \text{ob } D_{ctf}(A[G]^Y)$. Then, for every $g \in G$,

$$f_*^{gc} \text{Tr}_A(gu) = |Z_g| \text{Tr}_{A[G]}(f_* u)(g^{-1}), \quad (6.12.1)$$

where

$$f_*^{gc} : H^0(X^{gc}, K_{X^{gc}}) \longrightarrow H^0(Y^d, K_{Y^d})$$

is the Gysin morphism defined by the adjunction map $f_*^{gc} K_{X^{gc}} = f_*^{gc} f^{gc!} K_{Y^d} \rightarrow K_{Y^d}$.

Indeed, applying 6.10.5 to f_*u and using 6.11.5 gives

$$\mathrm{Tr}_A(f_*(gu)) = |Z_g| \mathrm{Tr}_{A[G]}(f_*u)(g^{-1}),$$

and 6.12.1 follows from the Lefschetz formula III, 4.5.1.

Let now $\ell \neq p$ be a prime. Put $\Lambda_n = \mathbb{Z}/\ell^n\mathbb{Z}$ for $n \geq 1$. Under the hypotheses of 6.11, let $L = (L_n)$ be a projective system of objects of $D_{ctf}(G, X, \Lambda_n)$ such that

$$L_m \otimes_{\Lambda_m}^L \Lambda_n \xrightarrow{\sim} L_n \quad (m \geq n),$$

and let $u = (u_n)$ be a projective system of equivariant correspondences $u_n \in \mathrm{Hom}(c_1^{*L_n}, c_2^{!L_n})$, compatible under extension of scalars. By compatibility of local traces with extension of scalars, the local terms

$$\mathrm{Tr}_{\Lambda_n}(gu_n) \in H^0(X^{gc}, K\Lambda_{n,X^{gc}})$$

define an element

$$\mathrm{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(X^{gc}, K_{\ell,X^{gc}}) := \varprojlim_n H^0(X^{gc}, K\Lambda_{n,X^{gc}}),$$

and the terms $f_*^{gc} \mathrm{Tr}_{\Lambda_n}(gu_n) = \mathrm{Tr}_{\Lambda_n}(gf_*u_n)$ define an element

$$f_* \mathrm{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(Y^d, K_{\ell,Y^d}) := \varprojlim_n H^0(Y^d, K\Lambda_{n,Y^d}).$$

Assume moreover that $f_*M_n \in \mathrm{ob} D_{ctf}(\Lambda_n[G]^Y)$ for all n . Then the local traces $\mathrm{Tr}_{\Lambda_n[G]}(f_*u_n)$ likewise define

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_*u) \in H^0(Y^d, K\mathbb{Z}_\ell[G]_{Y^d, \mathfrak{h}}) := \varprojlim_n H^0(Y^d, K\Lambda_n[G]_{Y^d, \mathfrak{h}}).$$

It follows from 6.12 that

$$f_*^{gc} \mathrm{Tr}_{\mathbb{Z}_\ell}(gu) = |Z_g| \mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_*u)(g^{-1}). \quad (6.13.1)$$

6.14–6.17. Open immersions and divisibility

Let X be a G - S -scheme, let $j : U \rightarrow X$ be an equivariant open immersion, and let $c : C \rightarrow X \times_S X$ be a morphism of G - S -schemes such that $c_2 = p_2c : C \rightarrow X$ is quasi-finite and of finite Tor-dimension. Assume that $c_1^{-1}(U) \subset c_2^{-1}(U)$. The trace morphism

$$\mathrm{Tr}_{c_2} : c_{2!}\Lambda_C \longrightarrow \Lambda_X$$

defines an equivariant filtered correspondence

$$\underline{c} \in \text{Hom}(c_1^* \Lambda_X, c_2^! \Lambda_X), \quad (6.14.2)$$

where Λ_X is endowed with the filtration defined by the submodule $i_! \Lambda_U$, namely $F^n \Lambda_X = \Lambda_X$ for $n \leq 0$, $F^1 \Lambda_X = i_! \Lambda_U$, and $F^n \Lambda_X = 0$ for $n \geq 1$. Its associated graded object consists of two equivariant correspondences

$$\text{gr}^0 \underline{c} = \underline{c}_{X-U,X} \in \text{Hom}(c_1^* \Lambda_{X-U,X}, c_2^! \Lambda_{X-U,X}), \quad \text{gr}^1 \underline{c} = \underline{c}_{U,X} \in \text{Hom}(c_1^* \Lambda_{U,X}, c_2^! \Lambda_{X-U,X}), \quad (6.14.3)$$

where $\Lambda_{X-U,X} = j_* \Lambda_{X-U}$ and $\Lambda_{U,X} = i_! \Lambda_U$. Moreover

$$\underline{c}_{X-U,U} = j_* \underline{c}_{X-U}, \quad (6.14.4)$$

where $\underline{c}_{X-U}$ is the correspondence defined by $\text{Tr}_{c_2, X-U}$.

Forgetting the action of G , one has local terms

$$\text{Tr}_\Lambda(\underline{c}), \quad \text{Tr}_\Lambda(\underline{c}_{U,X}), \quad \text{Tr}_\Lambda(\underline{c}_{X-U,X}) \in H^0(X^c, K_{X^c}), \quad (6.14.5)$$

which, by additivity of traces (III, 4.13.1), satisfy

$$\text{Tr}_\Lambda(\underline{c}) = \text{Tr}_\Lambda(\underline{c}_{U,X}) + \text{Tr}_\Lambda(\underline{c}_{X-U,X}). \quad (6.14.6)$$

Furthermore, by 6.14.4,

$$\text{Tr}_\Lambda(\underline{c}_{X-U,X}) = j_*^c \text{Tr}_\Lambda(\underline{c}_{X-U}), \quad (6.14.7)$$

where j_*^c is the direct-image morphism attached to $j^c : (X - U)^c \rightarrow X^c$.

Recall that if X is smooth, c is a closed immersion, and X^c is finite, then $\text{Tr}_\Lambda(\underline{c})$ is the image in Λ of the intersection multiplicity $(C \cdot X)$ of C with the diagonal. If moreover $X - U$ is smooth, then $\text{Tr}_\Lambda(\underline{c}_{X-U})$ is the image in Λ of $(c_2^{-1}(X - U) \cdot (X - U))$. Also, if $X - U$ is proper over S and $c|_{X-U}$ is the diagonal, the Lefschetz formula III, 4.7 gives that

$$\int_{X-U} \text{Tr}(\underline{c}_{X-U}) \in \Lambda$$

is the Euler-Poincaré characteristic of $X - U$ with values in Λ .

Now let a diagram 6.11.1 be given, with f, c, d proper, and let $U \rightarrow X$ be an equivariant open immersion. Assume that c_2 is quasi-finite and of finite Tor-dimension, that $c_1^{-1}(U) \subset c_2^{-1}(U)$, that G acts trivially on C and D , and that

$$f|_U : U \longrightarrow f(U) = V$$

is a Galois étale covering with group G . This last hypothesis implies that $Rf_*(\Lambda_{U,X}) \in \text{ob } D_{ctf}(A[G]^Y)$; indeed it is concentrated in degree zero and is the extension by zero of a sheaf on V locally isomorphic to $\Lambda[G]$. Thus one can apply 6.12 to $L = \Lambda_{U,X}$ and $u = \underline{c}_{U,X}$, and, using 6.14.6 and 6.14.7, one gets:

Proposition 6.16. Under the hypotheses of 6.15, for every $g \in G$,

$$f_*^{gc} \text{Tr}_\Lambda(\underline{gc}) - (fj)_*^{gc} \text{Tr}_\Lambda(\underline{gc}_{X-U}) = |Z_g| \text{Tr}_{A[G]}(f_* \underline{c}_{U,X})(g^{-1}), \quad (6.16.1)$$

where $j : X - U \rightarrow X$ is the inclusion and Z_g is the centralizer of g .

Applying 6.16 to the projective system of correspondences $\underline{c}_\ell = (\underline{c}^{\Lambda_n})$, with $\Lambda_n = \mathbb{Z}/\ell^n$ and $\ell \neq p$, and passing to the limit in 6.14.6 and 6.14.7, gives:

Corollary 6.17. Under the hypotheses of 6.15, for every $g \in G$,

$$f_*^{gc} \text{Tr}_{\mathbb{Z}_\ell}(\underline{gc}_\ell) - (fj)_*^{gc} \text{Tr}_{\mathbb{Z}_\ell}(\underline{gc}_{\ell,X-U}) = |Z_g| \text{Tr}_{\mathbb{Z}_\ell[G]}(f_* \underline{c}_{\ell,U,X})(g^{-1}), \quad (6.17.1)$$

with the notation of 6.16.

Taking into account compatibility of local traces with localization and the remarks made in 6.14, one sees that 6.17 gives an affirmative answer, in a more general formulation, to Grothendieck's divisibility conjecture (XII, 4.5) in the case where the endomorphism φ of G considered there is the identity.

In fact, inspired by 5.13, it is not difficult to formulate analogous divisibility statements to 6.12, 6.13, 6.16, and 6.17 in the case where c is not G -equivariant but c_2 is, and c_1 satisfies

$$c_1(xg) = c_1(x)\varphi(g)$$

for every point x of G , with a fixed endomorphism φ of G . The formulae then read as above, with Z_g replaced by the φ -centralizer of g^{-1} (5.13.5).

6.18–6.24. Invariants and the comparison with Grothendieck local terms

Put

$$N_G = \sum_{g \in G} g \in \Lambda[G]. \quad (6.18.1)$$

Let X be an S -scheme. For $L \in \text{ob } D(A[G]^X)$, left multiplication by N_G induces a morphism in $D(X)$

$$N_G : \Lambda \overset{L}{\otimes}_{A[G]} L \longrightarrow \text{RHom}_{A[G]}(\Lambda, L), \quad (6.18.2)$$

where Λ is regarded as a $\Lambda[G]$ -algebra through the natural augmentation sending every g to 1.

Lemma 6.18.3. If $L \in \text{ob } D_{ctf}(A[G]^X)$, the morphism 6.18.2 is an isomorphism.

By dévissage one is reduced to the case in which L is perfect, then to $L = \Lambda[G]$. The assertion then follows from the fact that N_G is a basis of the invariant module of the left $\Lambda[G]$ -module $\Lambda[G]$.

Proposition 6.19. In the situation of 6.11, let $M \in \text{ob } D_{ctf}(Y)$, let $v \in \text{Hom}(d_1^{*M}, d_2^{!M})$, and let $a : M \rightarrow f_*L$ be a morphism in $D(A[G]^Y)$, where M is regarded as an object of $D(A[G]^Y)$ through the augmentation $\Lambda[G] \rightarrow \Lambda$. Assume that the square

$$\begin{array}{ccc} d_1^{*M} & \xrightarrow{v} & d_2^{!M} \\ \downarrow d_1^{*a} & & \downarrow d_2^{!a} \\ d_1^* f_* L & \xrightarrow{f_* u} & d_2^! f_* L \end{array}$$

is commutative. Assume moreover that $f_* M \in \text{ob } D_{ctf}(A[G]^Y)$ and that a induces an isomorphism

$$M \xrightarrow{\sim} \text{RHom}_{A[G]}(\Lambda, f_* L).$$

Then

$$\text{Tr}_\Lambda(v) = \sum_{g \in G_{\mathfrak{h}}} \text{Tr}_{A[G]}(f_* u)(g^{-1}). \quad (6.19.1)$$

Indeed, by 6.18.3, composing a with N_G^{-1} gives an isomorphism

$$M \xrightarrow{\sim} \Lambda \overset{L}{\otimes}_{A[G]} f_* L$$

under which v is identified with $\Lambda \overset{L}{\otimes}_{A[G]} f_* u$. By 6.7.2, $\text{Tr}_\Lambda(v)$ is the image of $\text{Tr}_{A[G]}(f_* u)$ under the map induced by the augmentation $\Lambda[G] \rightarrow \Lambda$. This map sends each conjugacy class to 1, which proves the formula.

Let X_i for $i = 1, 2$ be G - S -schemes, put $X = X_1 \times_S X_2$, and let $c : C \rightarrow X$ be a G - S -morphism such that c_2 is quasi-finite and of finite Tor-dimension. For $L_i \in \text{ob } D^+(G, X_i)$, the trace morphism $\text{Tr}_{c_2} : c_{2!} c_2^{*L_2} \rightarrow L_2$ defines by adjunction a map, again denoted

$$\text{Tr}_{c_2} : c_2^{*L_2} \longrightarrow c_2^{!L_2}, \quad (6.20.1)$$

and hence every equivariant morphism $u \in \text{Hom}(c_1^{*L_1}, c_2^{*L_2})$ determines an equivariant correspondence

$$u^! = \text{Tr}_{c_2} \circ u \in \text{Hom}(c_1^{*L_1}, c_2^{*L_2}). \quad (6.20.2)$$

Proposition 6.21. Under the hypotheses of 6.11, assume moreover that an open immersion $i : V \hookrightarrow Y$ is given such that $(fc_1)^{-1}(V)$ is connected, that $d_1^{-1}(V) \subset d_2^{-1}(V)$, and that f induces a principal Galois covering with group G ,

$$U = f^{-1}(V) \xrightarrow{f_V} V.$$

Assume also that $M = i_! E$ with $E \in \text{ob } D(V)$, and that one has an isomorphism

$$\alpha : f_V^* E \xrightarrow{\sim} \Lambda_X \otimes_\Lambda E_0$$

in $D(G, X)$, with $E_0 \in \text{ob } D(\Lambda[G])$ perfect over Λ . Let $v \in \text{Hom}(d_1^{*M}, d_2^{*M})$ be as above, and let $v_0 \in \text{Hom}_{\Lambda[G]}(E_0, E_0)$ be the endomorphism defined, via α , by $f^*v|_{c_1^{-1}(U)}$. If $\underline{c}_{U,X}$ denotes the correspondence of 6.14.3, then, with the notation 6.20.2,

$$\text{Tr}_\Lambda(v^!) = \sum_{g \in G_{\mathfrak{q}}} \text{Tr}_{A[G]}(f_* \underline{c}_{U,X})(g^{-1}) \text{Tr}_\Lambda(gv_0). \quad (6.23.1)$$

Because of the central character of Tr_Λ , the right hand side does not depend on the choice of α . The projection formula $f_* f^* \Lambda_Y \otimes M \xrightarrow{\sim} f_* f^* M$ and 6.15 show that M satisfies the hypotheses needed above. The isomorphism α identifies $(f^*v)^!$ with the external tensor product $\underline{c}_{U,X} \otimes v_0$. Thus $f_* f^* M$ is identified with $f_*(\Lambda_{U,X}) \otimes E_0$, and $f_*(f^*v)^!$ with $f_* \underline{c}_{U,X} \otimes v_0$. For $g \in G$, the Z_g -equivariant correspondence $gf_*(f^*v)^!$ is identified with $gf_* \underline{c}_{U,X} \otimes gv_0$. Since 6.10.4 gives

$$\text{Tr}_{A[G]}(f_*(f^*v)^!)(g^{-1}) = \text{Tr}_{A[Z_g]}(gf_*(f^*v)^!)(e),$$

it is enough, using 6.22.1, to prove

$$\text{Tr}_{A[Z_g]}(gf_* \underline{c}_{U,X} \otimes gv_0)(e) = \text{Tr}_{A[G]}(f_* \underline{c}_{U,X})(g^{-1}) \text{Tr}_\Lambda(gv_0),$$

or equivalently, by 6.10.4,

$$\text{Tr}_{A[Z_g]}(gf_* \underline{c}_{U,X} \otimes gv_0)(e) = \text{Tr}_{A[Z_g]}(gf_* \underline{c}_{U,X})(e) \text{Tr}_\Lambda(gv_0). \quad (6.23.2)$$

Replacing G by Z_g , one may suppose $g = e$, and then, returning to the definition of traces, it suffices to establish the following lemma.

Lemma 6.23.3. Let P be a G_V -module locally a direct factor of a free module of finite

type, put $P' = i_! P$, and let Q be a complex of $\Lambda[G]$ -modules perfect over Λ . Then $P \otimes_{\Lambda}^L Q$, with the diagonal action of G , is perfect over $\Lambda[G]$, and the square

$$\begin{array}{ccc} \delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^{*P'}, p_2^{!P'}) \otimes^L \mathrm{RHom}_{\Lambda[G]}(Q, Q) & \xrightarrow{(1)} & K_Y \\ \downarrow (*) & & \parallel \\ \delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^{*P'} \otimes^L Q, p_2^{!P'} \otimes^L Q) & \xrightarrow{(3)} & K_Y \end{array}$$

is commutative. Here $\delta : Y \rightarrow Y \times_S Y$ is the diagonal; (1) is the tensor product of

$$\delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^{*P'}, p_2^{!P'}) \xrightarrow{6.5.3} K\Lambda[G]_{Y, \natural} \xrightarrow{x \mapsto x(e)} K_Y$$

and

$$\mathrm{RHom}_{\Lambda[G]}(Q, Q) \xrightarrow{\mathrm{Tr}_{\Lambda}} \Lambda;$$

(*) is induced from the external tensor product by restriction along the diagonal $\Lambda[G] \rightarrow \Lambda[G] \otimes \Lambda[G]$; and (3) is 6.5.3 followed by $x \mapsto x(e)$.

The first assertion is clear: one is reduced to $P = \Lambda[G]_V$, and then $P \otimes_{\Lambda}^L Q$ is the tensor product of P by Q with trivial G -action. For the second assertion, 6.5.4 and 6.5.5 give

$$\delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^{*P'}, p_2^{!P'}) \xrightarrow{\sim} D_{\Lambda[G]}(i_! P) \otimes_{\Lambda[G]}^L i_! P \xrightarrow{\sim} i_!(D_{\Lambda[G]} P \otimes_{\Lambda[G]}^L P),$$

so one is reduced to checking the induced square on V ; thus one may assume $V = Y$. The square becomes

$$\begin{array}{ccc} \mathrm{RHom}_{\Lambda[G]}(P, K_Y \otimes^L P) \otimes^L \mathrm{RHom}_{\Lambda[G]}(Q, Q) & \xrightarrow{(1)} & K_Y \\ \downarrow & & \parallel \\ \mathrm{RHom}_{\Lambda[G]}(P \otimes^L Q, K_Y \otimes^L P \otimes^L Q) & \xrightarrow{(3)} & K_Y. \end{array}$$

Here (1) is given by $\mathrm{Tr}_{\Lambda[G]}(-)(e) \otimes \mathrm{Tr}_{\Lambda}$, and (3) by $\mathrm{Tr}_{\Lambda[G]}(-)(e)$. By descent one is reduced to $P = \Lambda[G]_Y$, and then one may suppose that G acts trivially on Q . The assertion is then an immediate consequence of 5.12.5.

If Λ is killed by a power of a prime $\ell \neq p$, the right hand side of 6.23.1 may be rewritten as

$$\sum_{g \in G_{\natural}} \mathrm{Tr}_{\mathbb{Z}_{\ell}[G]}(f_{*} \mathcal{C}_{\ell, U, X})(g^{-1}) \mathrm{Tr}_{\Lambda}(g v_0),$$

with the notation of 6.17. Formula 6.17.1 shows that Grothendieck's local terms in XII, 6.2 coincide with Verdier's local terms, at least when the endomorphism φ of G considered

there is the identity; as already observed, this restriction is not serious. If Y is proper over S , combining 6.23.1 with the Lefschetz formula for $Y \rightarrow S$ (III, 4.7) gives a formula generalizing XII, 6.3, and hence in particular the Lefschetz formula for Frobenius proved in SGA 4 $\frac{1}{2}$, *Rapport*, and in Part I of the present exposé.

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SGA 4 $\frac{1}{2}$: *Cohomologie étale*, by P. Deligne, Lecture Notes in Mathematics 569, Springer, 1977. In SGA 4 $\frac{1}{2}$: *Th. Finitude* refers to the finiteness theorems in ℓ -adic cohomology; *Cycle* to the cohomology class associated with a cycle; and *Rapport* to the report on the trace formula.

Exposé V. J -adic projective systems

J. P. Jouanolou

1. Generalities on abelian A -categories

1.1. Definitions and first properties

Let A be a commutative ring. An A -category $\mathcal{C}$ is a category for which, for each pair (X, Y) of objects, the set $\text{Hom}(X, Y)$ is endowed with an A -module structure, compatibly with composition. Thus for three objects X, Y, Z the composition map

$$\text{Hom}(Y, Z) \times \text{Hom}(X, Y) \longrightarrow \text{Hom}(X, Z)$$

is A -bilinear.

An *abelian A -category* is an additive A -category in which every morphism has a kernel and a cokernel and in which every morphism factors canonically as an epimorphism followed by a monomorphism. More precisely, one assumes an equivalence between $\mathcal{C}$ and a full subcategory of the category of sheaves of A -modules on a site, so that finite inverse and direct limits, kernels, cokernels, images, and coimages are representable and are computed sheafwise.

An A -*functor*, respectively an additive A -functor, between A -categories is a functor, respectively an additive functor, whose action on Hom -sets is A -linear. An (A, B) -*functor* from an A -category to a B -category consists of a ring homomorphism $A \rightarrow B$ and a B -functor to the target. When $A = B$ and the map is the identity, this is simply an A -functor.

Let $\mathcal{C}$ be an abelian A -category. An object M of $\mathcal{C}$ is said to be of *finite type* (respectively *noetherian*) if, for every filtered inductive system $(N_i)_{i \in I}$ of subobjects of M , the canonical morphism

$$\varinjlim_i N_i \longrightarrow M$$

has the expected finiteness behavior: in the finite-type case the union stabilizes as soon as it is all of M , and in the noetherian case every increasing sequence of subobjects is stationary. An object is *coherent* if it is of finite type and if, for every map $u : N \rightarrow M$ with N of finite type, $\text{Ker}(u)$ is of finite type. For ordinary modules these are the usual notions.

If J is an ideal of A , the *canonical J -filtration* of an object M is the decreasing filtration by the images $J^n M$ of $J^n \otimes_A M \rightarrow M$. An object is called *J -adic* if this filtration is J -adic: for every n , the canonical map $J \otimes_A J^n M \rightarrow J^{n+1} M$ is an epimorphism and the transition

maps $M/J^{n+1}M \rightarrow M/J^nM$ are epimorphisms.

Throughout the exposé, the ideal J is assumed to satisfy the Artin–Rees condition (AR) of 1.1.8(ii). We shall use without further comment the elementary facts that the full subcategory of noetherian objects is thick and abelian: it is stable under subobjects, quotients, and extensions.

1.2–1.3. Projective systems and AR-zero systems

Let $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$ denote the category of projective systems indexed by $\mathbb{N}$ with values in $\mathcal{C}$. A system $X = (X_n)$ is called *AR-zero* if for every n there is an integer $m \geq n$ such that the transition map $X_m \rightarrow X_n$ is zero. The AR-zero systems form a thick subcategory, and in fact a localizing subcategory, of $\text{Hom}(\mathbb{N}^\circ, \mathcal{C})$ in the sense of Gabriel.

The quotient category

$$\text{Hom}(\mathbb{N}^\circ, \mathcal{C}) / \text{Hom}(\mathbb{N}^\circ, \mathcal{C})_{\text{AR-zero}}$$

is therefore abelian, and the canonical functor is exact. If $\mathcal{C}'$ is a full thick abelian subcategory of $\mathcal{C}$, one writes $\text{Hom}_{\text{AR}}(\mathbb{N}^\circ, \mathcal{C}')$ for the full subcategory of systems which are AR-isomorphic to systems with values in $\mathcal{C}'$. The usual properties of quotient categories then give the corresponding thick abelian subcategory in the quotient.

1.4. Examples and constructions

For an object M of $\mathcal{C}$, let $\kappa(M)$ denote the constant projective system with value M . The functor

$$\kappa : \mathcal{C} \longrightarrow \text{Hom}(\mathbb{N}^\circ, \mathcal{C})$$

is fully faithful and exact.

For an integer $r \geq 0$, the translation functor $[r]$ on projective systems is defined by

$$(X[r])_n = X_{n+r}.$$

It is exact, and there is a canonical monomorphism $X[r] \rightarrow X$ whose cokernel is AR-zero.

If $\mathcal{C}$ is complete, the projective limit

$$\varprojlim X = \varprojlim_n X_n$$

is an exact-left functor. For ordinary countable systems one has the standard exact sequence involving $R^1 \varprojlim$, and the higher derived functors vanish under the hypotheses used below.

The separated J -adic completion of an object M is

$$\widehat{M} = \varprojlim_n M/J^n M$$

when it exists. If $\mathcal{C}$ is complete and countable products are exact, completion is exact on noetherian objects. A projective system X is *strict J -adic* if X_n is killed by J^{n+1} , the transition maps are epimorphisms, and $X_m/J^{n+1}X_m \xrightarrow{\sim} X_n$ for $m \geq n$. Such a system is noetherian, respectively artinian, if its components are noetherian, respectively artinian.

The following elementary lemmas will be used repeatedly. A noetherian object has all subobjects noetherian. If M is separated and complete for the J -adic topology, then the system $(M/J^n M)_n$ is strict J -adic, and its image in the quotient by AR-zero systems is isomorphic to the image of the constant system $\kappa(M)$.

2. Mittag-Leffler–Artin–Rees and calculus of fractions

In this section $\mathcal{C}$ is an abelian A -category and $J \subset A$ satisfies (AR). A projective system $X = (X_n)$ satisfies the *Mittag-Leffler–Artin–Rees condition*, abbreviated (MLAR), if for every n there is $m \geq n$ such that for all $p \geq m$ the image of $X_p \rightarrow X_n$ is equal to the image of $X_m \rightarrow X_n$. In other words, the family of images

$$\mathrm{Im}(X_m \rightarrow X_n)_{m \geq n}$$

is stationary. A stricter version asks in addition that this stable image be a J -adic object of $\mathcal{C}$.

A system satisfying (MLAR) is AR-isomorphic to a strict projective system whose transition maps are epimorphisms. The condition is stable under short exact sequences in the familiar way. A J -adic projective system satisfies (MLAR) if and only if it is strict. Every noetherian projective system satisfies (MLAR).

Let X be a strict J -adic system. Its kernel is

$$\mathrm{Ker}(X) = (\mathrm{Ker}(X_n \rightarrow X_0))_n.$$

This is a strict subsystem, and the quotient is the constant system $\kappa(X_0)$. If Y is a strict subsystem of X , we say that Y is *J -adapted* to X if for every n ,

$$Y_n = X_n \cap Y_0 \quad \text{inside } X_0.$$

For such a subsystem, Y is J -adic exactly when the associated graded system is a module over $\mathrm{gr}_J(A)$ in the evident sense.

The associated graded object of a strict J -adic system is

$$\mathrm{gr}(X) = \bigoplus_{n \geq 0} X_n / X_{n+1},$$

viewed as a graded module over $\mathrm{gr}_J(A)$. The system X is noetherian, respectively artinian, if and only if $\mathrm{gr}(X)$ is noetherian, respectively artinian, as a graded $\mathrm{gr}_J(A)$ -module.

Localizing subcategories in the sense of Gabriel will be used to express the calculus of fractions. If $\mathcal{L}$ is a localizing subcategory of an abelian category $\mathcal{A}$, a morphism is a $\mathcal{L}$ -equivalence if its kernel and cokernel belong to $\mathcal{L}$. The quotient $\mathcal{A}/\mathcal{L}$ is the localization with respect to these equivalences, and the canonical functor $\mathcal{A} \rightarrow \mathcal{A}/\mathcal{L}$ is exact and universal for exact functors annihilating $\mathcal{L}$.

Applied to the localizing subcategory of AR-zero systems, this gives the category in which AR-isomorphisms are inverted. In particular, for a strict J -adic system X , the functor sending a J -adapted strict subsystem $Y \subset X$ to its image in the quotient is an equivalence from the category of such subsystems to the category of subobjects of the image of X in the quotient.

3. J -adic and AR- J -adic projective systems

A projective system is called *J -adic* if it is isomorphic to a strict J -adic system. A system is called *AR- J -adic* if it is AR-isomorphic to a strict J -adic system. We write

$$J\text{-adn}(\mathcal{C}), \quad J\text{-adf}(\mathcal{C}), \quad \mathrm{AR}\text{-}J\text{-adn}(\mathcal{C}), \quad \mathrm{AR}\text{-}J\text{-adf}(\mathcal{C})$$

for the corresponding noetherian and artinian subcategories.

A J -adic system remains J -adic after any translation $[r]$. Moreover, every AR- J -adic noetherian system is represented, in the quotient by AR-zero systems, by a translate $Y[r]$ of an actual J -adic noetherian system Y .

Assume now that $\mathcal{C}$ is complete and that countable products are exact. For any projective system X , the functor $\varprojlim$ is left exact, and its derived functors satisfy

$$R^q \varprojlim X = 0 \quad (q \geq 2).$$

If X is AR-zero, then $R^q \varprojlim X = 0$ for all $q \geq 0$. If X is AR- J -adic noetherian, then

$$R^1 \varprojlim X = 0.$$

Consequently $\varprojlim X$ is separated and complete for the J -adic topology, and the canonical

morphisms

$$X_n \longrightarrow \varprojlim X/J^{n+1} \varprojlim X$$

are isomorphisms.

The J -adic completion functor

$$\Lambda_J(M) = \varprojlim_n M/J^n M$$

is additive and left exact. Its left derived functors satisfy

$$L_i \Lambda_J(M) = 0 \quad (i \geq 2).$$

For M noetherian, one has $L_1 \Lambda_J(M) = 0$, and $\Lambda_J(M) \cong \varprojlim_n M/J^n M$. If $\mathcal{C}$ is complete with exact countable products and M is noetherian, the canonical map

$$\hat{A} \otimes_A M \longrightarrow \Lambda_J(M)$$

is an isomorphism; this is the familiar exactness of completion for finite modules in this categorical setting.

4. Filtrations and gradings

Let $X = (X_n)$ be a projective system. A filtration of X is a compatible family of decreasing filtrations $F^\bullet X_n$. It is called J -good if, for every n , the filtration on X_n is J -good, i.e. for some n_0 ,

$$J^p F^{n_0} X_n = F^{n_0+p} X_n \quad (p \geq 0).$$

The associated graded system is

$$\mathrm{gr}(X) = (\mathrm{gr}(X_n))_n, \quad \mathrm{gr}(X_n) = \bigoplus_{p \geq 0} F^p X_n / F^{p+1} X_n.$$

If X is strict J -adic and has a J -good filtration, then $\mathrm{gr}(X)$ is a strict $\mathrm{gr}_J(A)$ -adic projective system. If moreover X is noetherian, so is $\mathrm{gr}(X)$.

The Artin–Rees lemma takes the following form. Let M be a noetherian object of an abelian A -category and $N \subset M$ a subobject. Give M its canonical J -filtration. Then the induced filtration on N is J -adic: for every n there exists m such that

$$N \cap J^m M \subset J^n N.$$

The proof is by applying (MLAR) to the projective system $(N/(N \cap J^n M))_n$.

For a noetherian object M , the canonical map

$$J^n \otimes_A M \longrightarrow J^n M$$

is an epimorphism. Moreover, the system $(J^n M)_n$ is AR-zero if and only if M is killed by a power of J . If an AR- J -adic noetherian system carries a J -good filtration, its associated graded object is AR- $\text{gr}_J(A)$ -adic and noetherian. In a short exact sequence of J -adic noetherian systems with J -good filtrations, the associated sequence of graded systems is exact. Finally, for a noetherian strict J -adic system X , there exists $r \geq 0$ such that, for every strict subsystem $Y \subset X$, the graded object $\text{gr}(Y[r])$ is a graded $\text{gr}_J(A)$ -module.

The Nakayama lemma also has the expected form. If M is separated and complete for the J -adic topology and $M/JM = 0$, then $M = 0$. Hence a morphism between separated and complete objects is an epimorphism if and only if it is so modulo J . Similarly, M is of finite type or noetherian if and only if M/JM has the corresponding property over A/J , with the usual finiteness condition on the kernels of $J^n \otimes_A M \rightarrow J^n M$ in the noetherian criterion.

5. Noetherian AR- J -adic systems

In this section $\mathcal{C}$ is an abelian A -category, J is an ideal satisfying (AR), and $\mathcal{C}$ has countable direct sums.

A projective system $X = (X_n)$ is *AR- J -adic noetherian* (respectively *artinian*) if it is AR- J -adic and its components are noetherian (respectively artinian). We write

$$\text{AR-}J\text{-adn}(\mathcal{C}), \quad \text{AR-}J\text{-adf}(\mathcal{C})$$

for the corresponding full subcategories of the AR quotient. The canonical functors

$$p_{\text{AR}}^n : J\text{-adn}(\mathcal{C}) \rightarrow \text{AR-}J\text{-adn}(\mathcal{C}), \quad p_{\text{AR}}^f : J\text{-adf}(\mathcal{C}) \rightarrow \text{AR-}J\text{-adf}(\mathcal{C})$$

are equivalences of categories.

We shall need Hilbert's theorem in the following form. Let A be a commutative ring, $\mathcal{C}$ an abelian A -category, and M an object of finite type, respectively noetherian. For every finitely generated A -algebra $B = \bigoplus_{n \geq 0} B_n$ graded by $\mathbb{N}$, the graded $(B, \mathcal{C})$ -module $B \otimes_A M$ is of finite type, respectively noetherian. The proof reduces at once to $B = A[T]$, where submodules of $A[T] \otimes_A M$ correspond to increasing arrays of subobjects of M ; the assertion is then precisely the finite-type or noetherian condition on M .

If $\mathcal{C}$ has countable direct sums, the same conclusion holds for an arbitrary finitely generated A -algebra B , not necessarily graded. It follows in particular that if A is noetherian and

$J \subset A$ is an ideal, then, for every noetherian J -adic system X , the graded object $\mathrm{gr}(X)$ is a noetherian $(\mathrm{gr}_J(A), \mathcal{C})$ -module. Indeed $\mathrm{gr}_J(A)$ is a finitely generated A -algebra and there is an epimorphism

$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{gr}(X).$$

Let $\mathcal{E}_{\mathcal{C}}$ be the full subcategory of $\underline{\mathrm{Hom}}(\mathbb{N}^{\circ}, \mathcal{C})$ whose objects are AR- J -adic noetherian systems.

Proposition 5.2.1. The category $\mathcal{E}_{\mathcal{C}}$ is stable under kernels and cokernels in $\underline{\mathrm{Hom}}(\mathbb{N}^{\circ}, \mathcal{C})$. Thus $\mathcal{E}_{\mathcal{C}}$ is an abelian category and the inclusion

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \underline{\mathrm{Hom}}(\mathbb{N}^{\circ}, \mathcal{C})$$

is exact.

Let $u : X \rightarrow Y$ be a morphism in $\mathcal{E}_{\mathcal{C}}$. One proves successively that $\mathrm{Coker}(u)$, $\mathrm{Im}(u)$, and $\mathrm{Ker}(u)$ are AR- J -adic and noetherian. For the image one uses the following lemma.

Lemma 5.2.1.1. Let

$$0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$$

be exact in $\underline{\mathrm{Hom}}(\mathbb{N}^{\circ}, \mathcal{C})$. If M and N are AR- J -adic noetherian, then L is AR- J -adic noetherian.

It is enough to prove the AR- J -adic assertion. One reduces as in the proof of 3.2.4 to the case where M and N are strict, and even J -adic noetherian. Then 3.1.3(ii) shows that L is strict. It is adapted to J , and its graded object, as a graded subobject of $\mathrm{gr}(M)$, is noetherian. The assertion follows from the Artin–Rees lemma 4.2.6. Applying this first to $0 \rightarrow \mathrm{Im}(u) \rightarrow Y \rightarrow \mathrm{Coker}(u) \rightarrow 0$ and then to $0 \rightarrow \mathrm{Ker}(u) \rightarrow X \rightarrow \mathrm{Im}(u) \rightarrow 0$ gives the proposition.

Proposition 5.2.2. The objects of the abelian category $\mathrm{AR}\text{-}J\text{-adn}(\mathcal{C})$ are noetherian.

It is enough to show that, for a J -adic noetherian object X , any increasing sequence of subobjects in the AR category is stationary. Such a sequence may be represented by morphisms

$$v_n : Y^n \longrightarrow X$$

which are monomorphisms in the AR category and factor through one another there. The assertion that v_n factors through v_{n+1} means that for some r there is a morphism $f : Y^n[r] \rightarrow Y^{n+1}$ with

$$v_n \circ a = v_{n+1} \circ f,$$

where $a : Y^n[r] \rightarrow Y^n$ is the canonical strict morphism. Hence the image of v_n is contained in that of v_{n+1} . Replacing Y^n by the image, one is reduced to an increasing sequence of strict subsystems of X . Since $\text{gr}(X)$ is noetherian, the increasing sequence of the corresponding graded systems is stationary, and so is the original sequence.

In summary: $\mathcal{E}_{\mathcal{C}}$ is a thick abelian subcategory of $\underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{C})$; passage to the quotient by AR-zero systems induces an equivalence

$$\mathcal{E}_{\mathcal{C}} / (\mathcal{E}_{\mathcal{C}} \cap \underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{C})_{\text{AR-zero}}) \xrightarrow{\sim} \text{AR-}J\text{-adn}(\mathcal{C});$$

and the objects of $\text{AR-}J\text{-adn}(\mathcal{C})$ are noetherian.

Theorem 5.2.3, stated in this language, says that both $J\text{-adn}(\mathcal{C})$ and $\text{AR-}J\text{-adn}(\mathcal{C})$ are abelian and noetherian categories. For $\text{AR-}J\text{-adn}(\mathcal{C})$ this is precisely what has just been proved; the category $J\text{-adn}(\mathcal{C})$ is equivalent to it.

Proposition 5.2.4. Stability under extensions. Let

$$0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$$

be exact in $\underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for all n . Assume that Z is AR- J -adic and that X is AR- J -adic artinian. Then Y is AR- J -adic.

As in the proof of 3.2.4, one reduces to the case in which Y and Z are strict and Z is J -adic. Then X is strict. For some r the projective system

$$(X_{n+r} / J^{n+1}X_{n+r})_{n \in \mathbb{N}}$$

is J -adic artinian. Since X is a quotient of it and is AR-isomorphic to it, one is reduced to the following lemma.

Lemma 5.2.4.1. Let

$$0 \rightarrow T \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$$

be exact in $\underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for all n . If X is J -adic artinian, Z is J -adic, and T is AR-zero, then Y is AR- J -adic.

For fixed n , the evident system

$$(Y_{n+k} / J^{n+1}Y_{n+k})_{k \in \mathbb{N}}$$

is essentially constant. Indeed, let S_n^k be the kernel of

$$X_{n+k} / J^{n+1}X_{n+k} \longrightarrow Y_{n+k} / J^{n+1}Y_{n+k}.$$

In the exact commutative diagram

$$\begin{array}{ccccccccc}
0 & \rightarrow & S_n^k & \rightarrow & X_{n+k}/J^{n+1}X_{n+k} & \rightarrow & Y_{n+k}/J^{n+1}Y_{n+k} & \rightarrow & Z_{n+k}/J^{n+1}Z_{n+k} \rightarrow 0 \\
& & \downarrow & & \downarrow \wr & & \downarrow & & \downarrow \wr \\
0 & \rightarrow & T_n & \rightarrow & X_n & \rightarrow & Y_n & \rightarrow & Z_n \rightarrow 0,
\end{array}$$

the S_n^k are subobjects of the artinian object T_n and form a decreasing sequence. Hence they are eventually constant, which gives the essential constancy.

Define a system Y' by

$$Y'_n = \varprojlim_r Y_{n+r}/J^{n+1}Y_{n+r}.$$

Then Y' is J -adic. Passing to the limit gives an exact commutative diagram

$$\begin{array}{ccccccccc}
0 & \rightarrow & S & \rightarrow & X & \rightarrow & Y' & \rightarrow & Z \rightarrow 0 \\
& & \downarrow & & \downarrow \text{id} & & \downarrow & & \downarrow \text{id} \\
0 & \rightarrow & T & \rightarrow & X & \rightarrow & Y & \rightarrow & Z \rightarrow 0.
\end{array}$$

Since T is AR-zero, the snake lemma shows that $Y' \rightarrow Y$ is an AR-isomorphism. Thus Y is AR- J -adic.

Corollary 5.2.5. Let

$$0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$$

be exact in $\underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for all n . Assume that X is AR- J -adic artinian, respectively artinian and noetherian, and that Z is AR- J -adic artinian, respectively noetherian. Then Y is AR- J -adic artinian, respectively noetherian.

5.3. AR- J -adic noetherian systems and ∂ -functors

Let $\mathcal{C}$ and $\mathcal{D}$ be two abelian A -categories and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an additive functor. It extends termwise to an additive functor

$$F : \underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{C}) \rightarrow \underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{D})$$

which commutes with translations.

Proposition 5.3.1. Let $\mathcal{C}$ and $\mathcal{D}$ be abelian A -categories, and let $T = (T^i)_{i \in \mathbb{N}}$ be an exact A -linear ∂ -functor from $\mathcal{C}$ to $\mathcal{D}$. Assume:

- (i) for every noetherian object M of $\mathcal{C}$ killed by a power of J , and every n , the object $T^n(M)$ is noetherian;

- (ii) for every finitely generated graded A -algebra B killed by J , and every noetherian graded $(B, \mathcal{C})$ -module N , the graded $(B, \mathcal{D})$ -modules $T^n(N)$ are noetherian.

Then, for every AR- J -adic noetherian object X of $\underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, the objects $T^n(X)$ are AR- J -adic and noetherian.

Since the T^n commute with translations, it suffices to treat the case where X is J -adic noetherian. Then $\text{gr}_J(X)$ is a noetherian graded $\text{gr}_J(A)$ -module. By hypothesis (ii) and a variant of a corollary of Shih's theorem (Appendix A.3), for each n one has: $T^n(X)$ satisfies (MLAR), and $\text{gr}_J T^n(X)$ is a noetherian graded $(\text{gr}_J(A), \mathcal{D})$ -module. The Artin–Rees lemma 4.2.6 completes the proof.

It follows that the ∂ -functor

$$T = (T^i)_{i \in \mathbb{N}} : \underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{C}) \rightarrow \underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{D})$$

induces an exact A -linear ∂ -functor

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \mathcal{E}_{\mathcal{D}},$$

commuting with translations, and hence, after passing to the quotient, an exact A -linear ∂ -functor, again denoted (T^i) ,

$$\text{AR-}J\text{-adn}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adn}(\mathcal{D}).$$

Appendix. Shih's theorem

The purpose of this appendix is to prove Shih's theorem in the form used in the proof of 5.3.1. Since this form is more general than the one stated in EGA $\mathcal{O}_{\text{III}}$, a direct proof, adapted from that of loc. cit., is required.

Fix an integer $r_0 \geq 1$. The data, for every $r \geq r_0$, of a bigraded object $E_r = (E_r^{pq})_{p,q \in \mathbb{Z}}$, endowed with a differential d_r of bidegree $(r, 1 - r)$, together with isomorphisms

$$H(E_r) \xrightarrow{\sim} E_{r+1},$$

will be called a *spectral filter*. The notion of morphism of spectral filters is evident.

A.1. Spectral sequences attached to ∂ -functors

Let $\mathcal{C}$ be an abelian category and let X be an object of $\mathcal{C}$ with a decreasing filtration $(F^p X)_{p \in \mathbb{Z}}$. For every exact ∂ -functor from $\mathcal{C}$ to an abelian category $\mathcal{D}$, the method of exact couples gives a spectral sequence

$$\mathbb{E}(X) : E_1^{pq} = T^{p+q}(F^p X / F^{p+1} X) \Rightarrow E^n = T^n(X),$$

whose filtration on E^n is

$$F^p(E^n) = \text{Im}(T^n(F^p X) \rightarrow T^n(X)) = \text{Ker}(T^n(X) \rightarrow T^n(X/F^p X)).$$

If the filtration on X is finite, this spectral sequence is weakly convergent (EGA $\mathcal{O}_{\text{III}}$, 11.1.3).

The construction is functorial. If $f : X \rightarrow Y$ is a morphism of filtered objects, there is a morphism of spectral sequences

$$\mathbb{E}(f) : \mathbb{E}(X) \rightarrow \mathbb{E}(Y)$$

which is $T^{p+q}(\text{gr}^p f)$ on E_1^{pq} and $T^n(f)$ on the abutment. If $u : X \rightarrow X$ is an endomorphism with $u(F^p X) \subset F^{p+h} X$ for an integer h independent of p , then $\mathbb{E}(u)$ is an endomorphism of bidegree $(h, -h)$ of the spectral filter, commuting with the differentials.

A.2. The case of projective systems

Let $X = (X_n)_{n \in \mathbb{Z}}$ be a projective system of objects of $\mathcal{C}$, with $X_i = 0$ for $i < 0$. If T is an exact ∂ -functor from $\mathcal{C}$ to $\mathcal{D}$, it extends naturally to a ∂ -functor on $\underline{\text{Hom}}(\mathbb{Z}^\circ, \mathcal{C})$. Giving X its natural filtration as a projective system yields a spectral sequence

$$\mathbb{E}(X) = \mathbb{E}(X_n)_{n \in \mathbb{Z}},$$

where $\mathbb{E}(X_n)$ is the spectral sequence associated with T and the induced filtration on X_n .

If X is strict, the filtration of the abutment $E^n(X)$ in this spectral sequence is the natural filtration of the projective system. Indeed,

$$F^p(E^n(X)) = \text{Im}(T^n(F^p X) \rightarrow T^n(X)),$$

and $(F^p X)_k = X_k$ for $k \geq p$ and 0 for $k < p$.

Suppose a ring S acts on X compatibly with its natural filtration. Then, for each $r \geq 1$,

the bigraded object $E_r(X)$ is canonically a graded $(\mathrm{gr}^\bullet S, \mathcal{D})$ -module; the same holds for

$$\mathcal{E}_r^{(n)}(X) = (\mathcal{E}_r^{p,q}(X))_{p+q=n}.$$

Moreover S acts on $T^n(X)$, and whenever $T^n(X)$ satisfies the Mittag-Leffler condition, the strict graded object $\mathrm{gr}_S^\bullet T^n(X)$ is a graded $(\mathrm{gr}^\bullet S, \mathcal{D})$ -module.

Theorem A.3. Suppose that, for an integer n , the graded $(\mathrm{gr}^\bullet S, \mathcal{D})$ -modules

$$\mathcal{E}_1^{(n)}(X) = T^n(\mathrm{gr}_S^\bullet X), \quad \mathcal{E}_1^{(n-1)}(X) = T^{n-1}(\mathrm{gr}_S^\bullet X)$$

are noetherian. Then:

- (i) $T^n(X)$ satisfies (MLAR);
- (ii) $\mathrm{gr}_S^\bullet T^n(X)$ is a noetherian graded $(\mathrm{gr}^\bullet S, \mathcal{D})$ -module.

For every pair (m, r) one has

$$\mathcal{E}_r^{(m)}(X) = (\mathcal{E}_r^{p, m-p}(X))_{p \in \mathbb{N}},$$

a graded submodule of $\mathcal{E}_1^{(m)}(X)$. Thus, for fixed m , the $\mathcal{E}_r^{(m)}(X)$ form an increasing sequence indexed by r . Since $\mathcal{E}_1^{(n)}(X)$ and $\mathcal{E}_1^{(n-1)}(X)$ are noetherian, the corresponding sequences are stationary. Hence for some r_0 and all $r \geq r_0$,

$$\mathcal{E}_r^{(n)}(X) = \mathcal{E}_{r_0}^{(n)}(X), \quad \mathcal{E}_r^{(n-1)}(X) = \mathcal{E}_{r_0}^{(n-1)}(X).$$

An equivalent characterization of the Mittag-Leffler condition then shows that the spectral sequence $\mathbb{E}(X)$ is weakly convergent in degrees n and $n - 1$. It follows that the filtration of $E^n(X) = T^n(X)$ is separated and that $\mathcal{E}_\infty^{(n)}(X)$ is a subobject of the noetherian module $\mathcal{E}_{r_0}^{(n)}(X)$.

To prove that $T^n(X)$ satisfies (ML), it is enough to prove that for each p the sequence

$$\mathrm{Im}(T^n(F^p X) \rightarrow T^n(X/F^{p+r} X))_{r \geq 1}$$

is stationary. Weak convergence identifies these images, for r large, with the kernels of maps to $E_\infty^{(n)}(X)$; since these kernels lie in a noetherian object, they are stationary. Assertion (ii) follows because $\mathrm{gr}_S^\bullet T^n(X)$ is a subquotient of $\mathcal{E}_\infty^{(n)}(X)$.

A.4. Application to ∂ -functors

Let $\mathcal{C}$ and $\mathcal{D}$ be abelian A -categories, and let $T = (T^i)_{i \in \mathbb{N}}$ be an exact A -linear ∂ -functor from $\mathcal{C}$ to $\mathcal{D}$. Let X be an object of $\underline{\text{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, and let S be a graded A -algebra acting on X compatibly with the transition maps. Assume that X is AR- J -adic noetherian; that $T^n(M)$ is noetherian for each noetherian M killed by a power of J ; and that, for every finitely generated graded A -algebra B killed by J and every noetherian graded $(B, \mathcal{C})$ -module N , the graded $(B, \mathcal{D})$ -modules $T^n(N)$ are noetherian.

Theorem A.4.1. Under these hypotheses, the objects $T^n(X)$ are AR- J -adic and noetherian.

By translating one is reduced to the case where X is J -adic noetherian. Then $\text{gr}_J^\bullet(X)$ is a noetherian graded $(\text{gr}_J^\bullet(A), \mathcal{C})$ -module. Hypothesis (iii) makes $\mathcal{E}_1^{(n)}(X)$ and $\mathcal{E}_1^{(n-1)}(X)$ noetherian, and Theorem A.3 gives the conclusion. This completes Exposé V.

Exposé VI. ℓ -adic cohomology

1. Categories of constructible ℓ -adic sheaves

1.1. Generalities

Let X be a scheme and let ℓ be a prime number. We write

$$\mathbb{Z}_\ell\text{-fc}(X)$$

for the category of constructible $\mathbb{Z}_\ell$ -sheaves on X , defined in SGA 5, V. It is an abelian category. We write

$$\mathbb{Q}_\ell\text{-fc}(X)$$

for the category of constructible $\mathbb{Q}_\ell$ -sheaves, i.e. the quotient of the preceding category by the thick subcategory of torsion sheaves. We have also defined the categories

$$\text{AR-}\mathbb{Z}_\ell\text{-fc}(X), \quad \text{AR-}\mathbb{Q}_\ell\text{-fc}(X).$$

Recall that giving a constructible $\mathbb{Z}_\ell$ -sheaf $\mathcal{F}$ is equivalent to giving a projective system $(\mathcal{F}_n)_{n \in \mathbb{N}}$ of constructible $\mathbb{Z}/\ell^{n+1}\mathbb{Z}$ -modules on X such that, for every n , the transition morphism $\mathcal{F}_{n+1} \rightarrow \mathcal{F}_n$ factors through an isomorphism

$$\mathcal{F}_{n+1}/\ell^{n+1}\mathcal{F}_{n+1} \xrightarrow{\sim} \mathcal{F}_n.$$

For two such sheaves $\mathcal{F} = (\mathcal{F}_n)$ and $\mathcal{G} = (\mathcal{G}_n)$,

$$\text{Hom}(\mathcal{F}, \mathcal{G}) = \varprojlim_n \text{Hom}(\mathcal{F}_n, \mathcal{G}_n).$$

For each n , the canonical map

$$\text{Hom}(\mathcal{F}, \mathcal{G}) \longrightarrow \text{Hom}(\mathcal{F}_n, \mathcal{G}_n)$$

is surjective, and its kernel consists of morphisms which factor through a torsion object.

The ring $\mathbb{Z}_\ell$ acts in the evident way on every component of a constructible ℓ -adic sheaf, and hence on the sheaf itself. Every morphism is compatible with these actions. Thus the category of constructible ℓ -adic sheaves is canonically an abelian $\mathbb{Z}_\ell$ -category. We shall call these objects constructible $\mathbb{Z}_\ell$ -sheaves. This should not be confused with a module over the sheaf of rings $(\mathbb{Z}_\ell)_X$, a notion not used here. The category will also be denoted $\mathbb{Z}_\ell\text{-fc}(X)$.

1.2. Twisted constant constructible $\mathbb{Z}_\ell$ -sheaves

A constructible $\mathbb{Z}_\ell$ -sheaf

$$\mathcal{F} = (\mathcal{F}_n)_{n \in \mathbb{N}}$$

is called *twisted constant* if and only if all the components $\mathcal{F}_n$ are locally constant on X_{et} . The category of twisted constant constructible $\mathbb{Z}_\ell$ -sheaves is the full subcategory of $\mathbb{Z}_\ell\text{-fc}(X)$ generated by these objects.

Assume that ℓ is invertible on X . For each n , raising to the ℓ th power gives an endomorphism of $\mathbb{G}_m$, and the sheaf

$$\mathbb{Z}/\ell^{n+1}\mathbb{Z}(1)$$

is the sheaf $\mu_{\ell^{n+1}}$ of ℓ^{n+1} st roots of unity. The projective system of these sheaves defines a twisted constant $\mathbb{Z}_\ell$ -sheaf denoted $\mathbb{Z}_\ell(1)$.

Let $\mathcal{F} = (\mathcal{F}_n)$ be a twisted constant constructible $\mathbb{Z}_\ell$ -sheaf. For every geometric point $\bar{x}$ of X , the geometric fiber $\mathcal{F}_{\bar{x}}$ is a finite type $\mathbb{Z}_\ell$ -module endowed with a continuous action of the fundamental group $\pi_1(X, \bar{x})$. The fiber functor

$$\mathcal{F} \longmapsto \mathcal{F}_{\bar{x}}$$

is an equivalence between twisted constant constructible $\mathbb{Z}_\ell$ -sheaves on X and finite type $\mathbb{Z}_\ell$ -modules with continuous action of $\pi_1(X, \bar{x})$.

For a finite type $\mathbb{Z}_\ell$ -module M we write M_X for the corresponding twisted constant sheaf. In particular,

$$\mathbb{Z}_\ell(r) = \mathbb{Z}_\ell(1)^{\otimes r} \quad (r \in \mathbb{Z}),$$

where for $r < 0$ one sets $\mathbb{Z}_\ell(r) = \mathbb{Z}_\ell(-r)^\vee$. If $\mathcal{F}$ is a twisted constant $\mathbb{Z}_\ell$ -sheaf, define

$$H^i(X, \mathcal{F}) = \varprojlim_n H^i(X, \mathcal{F}_n).$$

These groups are finite type $\mathbb{Z}_\ell$ -modules when X is proper over a separably closed field.

1.3. Invertible ℓ -adic sheaves

Let $\mathcal{L}$ be a constructible $\mathbb{Z}_\ell$ -sheaf. We say that $\mathcal{L}$ is *invertible* if, for every n , the component $\mathcal{L}_n$ is an invertible $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X$ -module, i.e. locally free of rank one. Equivalently, $\mathcal{L}$ is locally free in the sense of the preceding paragraph and the geometric fibers of the $\mathcal{L}_n$ are non-zero cyclic abelian groups.

For an invertible $\mathcal{L}$ put

$$\mathcal{L}^{-1} = \mathcal{H}om(\mathcal{L}, \mathbb{Z}_\ell).$$

There is then an isomorphism

$$\mathcal{L} \otimes \mathcal{L}^{-1} \xrightarrow{\sim} \mathbb{Z}_\ell,$$

obtained componentwise. Following EGA $\mathcal{O}_I$, 5.4.4, set $\mathcal{L}^{\otimes 0} = \mathbb{Z}_\ell$ and, for $n \geq 1$, $\mathcal{L}^{\otimes(-n)} = (\mathcal{L}^{-1})^{\otimes n}$. With this notation there are canonical functorial isomorphisms

$$\mathcal{L}^{\otimes m} \otimes \mathcal{L}^{\otimes n} \xrightarrow{\sim} \mathcal{L}^{\otimes(m+n)}$$

for all integers m, n , satisfying the usual associativity rules. Thus $(\mathcal{L}^{\otimes i})_{i \in \mathbb{N}}$ and $(\mathcal{L}^{\otimes i})_{i \in \mathbb{Z}}$ are graded $\mathbb{Z}_\ell$ -algebras. The sheaf $\mathbb{Z}_\ell(1)$ is invertible, and the convention above gives $\mathbb{Z}_\ell(r) = \mathbb{Z}_\ell(1)^{\otimes r}$ for all $r \in \mathbb{Z}$.

1.4. Categories built from constructible $\mathbb{Z}_\ell$ -sheaves

Let A be a finite $\mathbb{Z}_\ell$ -algebra. Define the category

$$A\text{-fc}(X)$$

of constructible A -sheaves as follows. An object is a pair $(\mathcal{F}, u)$ consisting of a constructible $\mathbb{Z}_\ell$ -sheaf $\mathcal{F}$ and a homomorphism of $\mathbb{Z}_\ell$ -algebras

$$u : A \longrightarrow \text{End}(\mathcal{F}).$$

A morphism $(\mathcal{F}, u) \rightarrow (\mathcal{F}', u')$ is a morphism $f : \mathcal{F} \rightarrow \mathcal{F}'$ of constructible $\mathbb{Z}_\ell$ -sheaves such that, for every $a \in A$, the square

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{f} & \mathcal{F}' \\ \downarrow u(a) & & \downarrow u'(a) \\ \mathcal{F} & \xrightarrow{f} & \mathcal{F}' \end{array}$$

commutes. The category $A\text{-fc}(X)$ is abelian, and the forgetful functor to $\mathbb{Z}_\ell\text{-fc}(X)$ is exact and faithful. The dual of a constructible A -sheaf $\mathcal{F}$ is

$$\mathcal{F}^\vee = \mathcal{H}om(\mathcal{F}, A_X).$$

If $f : X \rightarrow Y$ is a morphism of schemes, inverse image and direct image extend in the evident way to the corresponding categories of constructible A -sheaves.

Let $p : \mathbb{Z}_\ell\text{-fc}(X) \rightarrow \mathbb{Q}_\ell\text{-fc}(X)$ be the quotient functor. For a constructible $\mathbb{Z}_\ell$ -sheaf $\mathcal{F}$, the

functor obtained by composing p with $\mathcal{G} \mapsto \mathcal{F} \otimes \mathcal{G}$ makes multiplication by non-zero elements of $\mathbb{Z}_\ell$ invertible, and hence defines a right-exact functor

$$\mathbb{Q}_\ell\text{-fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-fc}(X).$$

Varying the first argument as well gives a right-exact bifunctor, called tensor product,

$$\mathbb{Q}_\ell\text{-fc}(X) \times \mathbb{Q}_\ell\text{-fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-fc}(X).$$

If X is connected and $\mathcal{F}, \mathcal{G}$ are twisted constant constructible $\mathbb{Q}_\ell$ -sheaves corresponding to representations V, W of $\pi_1(X)$, then $\mathcal{F} \otimes \mathcal{G}$ is again locally constant and corresponds to the diagonal representation on $V \otimes W$.

1.5. Constructible ℓ -adic sheaves over the spectrum of a field

If k is separably closed, then $\mathbb{Z}_\ell\text{-fc}(\text{Spec } k)$ is identified with the category of finite type $\mathbb{Z}_\ell$ -modules, and $\mathbb{Q}_\ell\text{-fc}(\text{Spec } k)$ with finite-dimensional $\mathbb{Q}_\ell$ -vector spaces.

If k is a field and $\bar{k}$ is a separable closure, the geometric-fiber functor at $\text{Spec}(\bar{k}) \rightarrow \text{Spec}(k)$ gives an equivalence

$$\mathbb{Z}_\ell\text{-fc}(\text{Spec } k) \xrightarrow{\sim} \{\text{finite type } \mathbb{Z}_\ell\text{-modules with continuous action of } \text{Gal}(\bar{k}/k)\}.$$

After inverting torsion, this identifies $\mathbb{Q}_\ell\text{-fc}(\text{Spec } k)$ with the category of finite-dimensional continuous ℓ -adic representations of $\text{Gal}(\bar{k}/k)$.

If k is finite with q elements, let

$$F_k \in \text{Gal}(\bar{k}/k)$$

denote the geometric Frobenius. A continuous ℓ -adic representation of $\text{Gal}(\bar{k}/k)$ is determined by the action of F_k . If V is a finite-dimensional $\mathbb{Q}_\ell$ -vector space with continuous action of $\text{Gal}(\bar{k}/k)$, the eigenvalues of F_k on V are algebraic numbers whose inverses are integral over $\mathbb{Z}$.

Let

$$\mathbb{Z}_\ell\text{-modn}(\pi_1)$$

be the category of finite type $\mathbb{Z}_\ell$ -modules endowed with a continuous action of π_1 .

Lemma 1.2.4.2. The preceding projective-limit functor

$$\varprojlim_{\mathbb{N}} : (\ell\text{-}\mathbb{Z})\text{-ad}(\text{Abf}(\pi_1)) \longrightarrow \mathbb{Z}_\ell\text{-modn}(\pi_1)$$

is an equivalence of categories.

It is clearly essentially surjective: a finite type $\mathbb{Z}_\ell$ -module E with a continuous action of π_1 is obtained from the projective system

$$(E/\ell^{n+1}E)_{n \in \mathbb{N}}$$

with the evident actions of π_1 .

2. Formalism of ℓ -adic cohomology

2.1. Inverse image

Let $f : X \rightarrow Y$ be a morphism of locally noetherian preschemes. The inverse-image functor

$$f^* : \text{Ab}(Y) \longrightarrow \text{Ab}(X)$$

is exact and therefore extends in the evident way to an exact functor, denoted by the same symbol,

$$f^* : \underline{\text{Hom}}(\mathbb{N}^\circ, \text{Ab}(Y)) \longrightarrow \underline{\text{Hom}}(\mathbb{N}^\circ, \text{Ab}(X)).$$

It plainly carries constructible $\mathbb{Z}_\ell$ -sheaves to constructible $\mathbb{Z}_\ell$ -sheaves and locally AR-zero systems to locally AR-zero systems. With the notation of 1.5.4 it therefore induces inverse-image functors

$$\begin{aligned} f^* : \mathbb{Z}_\ell\text{-fc}(Y) &\longrightarrow \mathbb{Z}_\ell\text{-fc}(X), \\ f^* : \mathbb{Q}_\ell\text{-fc}(Y) &\longrightarrow \mathbb{Q}_\ell\text{-fc}(X). \end{aligned}$$

These functors are exact, commute with tensor products, and preserve twisted constant sheaves. If A is a finite $\mathbb{Z}_\ell$ -algebra, the same construction gives inverse-image functors for constructible A -sheaves.

2.2. Direct image

Let again $f : X \rightarrow Y$ be a morphism of locally noetherian preschemes. The direct-image functor on étale sheaves is left exact and so has right derived functors. Applying it term by term to a projective system gives a left exact functor

$$f_* : \underline{\text{Hom}}(\mathbb{N}^\circ, \text{Ab}(X)) \longrightarrow \underline{\text{Hom}}(\mathbb{N}^\circ, \text{Ab}(Y)).$$

The corresponding derived functors are denoted $R^i f_*$. When f is proper and of finite type, the finiteness theorem for étale cohomology shows that $R^i f_*$ carries constructible $\mathbb{Z}_\ell$ -sheaves

to constructible $\mathbb{Z}_\ell$ -sheaves. Thus one obtains functors

$$R^i f_* : \mathbb{Z}_\ell\text{-fc}(X) \longrightarrow \mathbb{Z}_\ell\text{-fc}(Y)$$

and, after passage to the quotient by torsion,

$$R^i f_* : \mathbb{Q}_\ell\text{-fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-fc}(Y).$$

They are compatible with formation of fibers in the usual sense: at a geometric point $\bar{y}$ of Y , the fiber of $R^i f_* \mathcal{F}$ is the i th cohomology of the geometric fiber $X_{\bar{y}}$ with coefficients in the pullback of $\mathcal{F}$.

2.3. Tensor products and internal Hom

The tensor product of projective systems is defined level by level. If $\mathcal{F} = (\mathcal{F}_n)$ and $\mathcal{G} = (\mathcal{G}_n)$ are constructible $\mathbb{Z}_\ell$ -sheaves, set

$$(\mathcal{F} \otimes \mathcal{G})_n = \mathcal{F}_n \otimes_{\mathbb{Z}/\ell^{n+1}\mathbb{Z}} \mathcal{G}_n.$$

This again defines a constructible $\mathbb{Z}_\ell$ -sheaf. The functor is right exact in each variable; its left derived functors are denoted $\text{Tor}_i^{\mathbb{Z}_\ell}(\mathcal{F}, \mathcal{G})$. For constructible $\mathbb{Q}_\ell$ -sheaves the tensor product is exact.

The internal Hom is defined similarly by

$$\mathcal{H}om(\mathcal{F}, \mathcal{G})_n = \mathcal{H}om_{\mathbb{Z}/\ell^{n+1}\mathbb{Z}}(\mathcal{F}_n, \mathcal{G}_n),$$

with the transition morphisms induced in the evident way. Its right derived functors are written $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$. The familiar adjunctions between tensor product and internal Hom remain valid in the ℓ -adic setting.

2.4. Cohomology

For a locally noetherian scheme X and a constructible $\mathbb{Z}_\ell$ -sheaf $\mathcal{F} = (\mathcal{F}_n)$, define

$$H^i(X, \mathcal{F}) = \varprojlim_n H^i(X, \mathcal{F}_n).$$

When X is proper over a separably closed field, these groups are finite $\mathbb{Z}_\ell$ -modules. Passing to $\mathbb{Q}_\ell$ gives

$$H^i(X, \mathcal{F} \otimes \mathbb{Q}_\ell) = H^i(X, \mathcal{F}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

If X is not proper, one uses cohomology with proper supports, defined by compactification. For separated morphisms of finite type, the construction is independent of the chosen compactification.

2.5. Finite cohomological dimension

Let X be of finite type over a separably closed field and let ℓ be invertible on X . Then X has finite ℓ -cohomological dimension. More precisely, if $\dim X = d$, then for every torsion constructible sheaf F killed by a power of ℓ , one has

$$H^i(X, F) = 0 \quad (i > 2d),$$

and similarly for cohomology with proper supports. This is the usual cohomological dimension theorem of SGA 4, XIV.

2.6. Cohomology with proper supports

Let $f : X \rightarrow S$ be a separated morphism of finite type between locally noetherian schemes, and let $\mathcal{F}$ be a constructible $\mathbb{Z}_\ell$ -sheaf on X . The cohomology with proper supports

$$H_c^i(X/S, \mathcal{F})$$

is the constructible $\mathbb{Z}_\ell$ -sheaf on S obtained by compactifying f and applying the corresponding direct image with extension by zero. If $j : U \rightarrow X$ is an open immersion and $i : Z \rightarrow X$ is the complementary closed immersion, then for every constructible $\mathbb{Z}_\ell$ -sheaf $\mathcal{F}$ on U there is the usual long exact localization sequence

$$\cdots \rightarrow R^i f_!(j_! \mathcal{F}) \rightarrow R^i f_!(Rj_* \mathcal{F}) \rightarrow R^i f_!(i_* i^* Rj_* \mathcal{F}) \rightarrow \cdots$$

3. ℓ -adic L - and Z -functions

3.1. Definitions

Let $\mathbb{F}_q$ be a finite field with q elements and let X be a scheme of finite type over $\mathbb{F}_q$. Let $|X|$ be the set of closed points of X . For $x \in |X|$, write $d(x)$ for the degree of the residue field $k(x)$ over $\mathbb{F}_q$, and put $N(x) = q^{d(x)}$.

Let $\mathcal{F}$ be a constructible $\mathbb{Q}_\ell$ -sheaf on X . For each closed point x , the geometric fiber $\mathcal{F}_{\bar{x}}$ is a finite-dimensional $\mathbb{Q}_\ell$ -vector space with a continuous action of $\text{Gal}(\overline{k(x)}/k(x))$. Let F_x be

geometric Frobenius. If the eigenvalues of F_x on $\mathcal{F}_{\bar{x}}$ are $\alpha_1, \dots, \alpha_n$, then

$$\det(1 - F_x t^{d(x)} \mid \mathcal{F}_{\bar{x}}) = \prod_{i=1}^n (1 - \alpha_i t^{d(x)}).$$

The associated Z -function is

$$Z(X, \mathcal{F}; t) = \prod_{x \in |X|} \frac{1}{\det(1 - F_x t^{d(x)} \mid \mathcal{F}_{\bar{x}})} \in \mathbb{Q}_\ell[[t]].$$

For $\mathcal{F} = \mathbb{Q}_\ell$ one writes simply $Z(X; t)$.

3.2. Lefschetz trace formula

Theorem 3.2.1. Lefschetz trace formula. Let X be proper and smooth over $\mathbb{F}_q$ and let $\mathcal{F}$ be a constructible $\mathbb{Q}_\ell$ -sheaf. Then

$$Z(X, \mathcal{F}; t) = \prod_{i=0}^{2 \dim X} \det(1 - F^* t \mid H^i(X \otimes \bar{\mathbb{F}}_q, \mathcal{F}))^{(-1)^{i+1}}.$$

The proof uses the Lefschetz formula for correspondences on proper smooth varieties, as developed in SGA 5, III, together with the Artin–Verdier theory of the L -functor. The essential point is that the global trace of Frobenius on cohomology is equal to the sum of its local terms at the fixed points, which are the rational points over finite extensions of $\mathbb{F}_q$.

3.3. The Weil conjectures

Let X be proper and smooth over a finite field $\mathbb{F}_q$, and let ℓ be different from the characteristic of $\mathbb{F}_q$. The Weil conjectures assert:

- (i) the characteristic polynomials

$$\det(1 - F^* t \mid H^i(X \otimes \bar{\mathbb{F}}_q, \mathbb{Q}_\ell))$$

have integral coefficients and are independent of ℓ ;

- (ii) the eigenvalues of F^* on $H^i(X \otimes \bar{\mathbb{F}}_q, \mathbb{Q}_\ell)$ are algebraic integers all of whose complex absolute values are $q^{i/2}$;

- (iii) Poincaré duality gives perfect pairings

$$H^i(X \otimes \bar{\mathbb{F}}_q, \mathbb{Q}_\ell) \times H^{2d-i}(X \otimes \bar{\mathbb{F}}_q, \mathbb{Q}_\ell(d)) \longrightarrow \mathbb{Q}_\ell.$$

These conjectures are proved by Deligne. The proof uses the trace formula and the theory of weights.

4. Cohomology of ℓ -adic completions

Let X be proper over $\mathrm{Spec}(\mathbb{Z}[1/\ell])$. For each $n \geq 1$, put

$$X_n = X \otimes_{\mathbb{Z}[1/\ell]} \mathbb{Z}[1/\ell, \zeta_{\ell^n}],$$

where ζ_{ℓ^n} is a primitive ℓ^n th root of unity. The cohomology groups

$$H^i(X_n, \mathbb{Z}/\ell^n \mathbb{Z})$$

form a projective system, and its limit

$$H_{\ell\text{-comp}}^i(X) = \varprojlim_n H^i(X_n, \mathbb{Z}/\ell^n \mathbb{Z})$$

is called the i th completed ℓ -adic cohomology group of X . These are finite type $\mathbb{Z}_\ell$ -modules and coincide, in the proper smooth case, with the usual ℓ -adic cohomology groups. With coefficients in a constructible $\mathbb{Z}_\ell$ -sheaf $\mathcal{F}$, define similarly

$$H_{\ell\text{-comp}}^i(X, \mathcal{F}) = \varprojlim_n H^i(X_n, \mathcal{F}/\ell^n \mathcal{F}).$$

There is also an Iwasawa spectral sequence. If $G = \mathrm{Gal}(\mathbb{Q}(\zeta_{\ell^\infty})/\mathbb{Q}) \cong \mathbb{Z}_\ell^\times$, then G acts on $H_{\ell\text{-comp}}^i(X)$ by transport of structure. These groups may therefore be regarded as modules over the Iwasawa algebra

$$\Lambda = \mathbb{Z}_\ell[[G]] = \varprojlim_n \mathbb{Z}_\ell[G/\ell^n \mathbb{Z}_\ell^\times].$$

The resulting Λ -modules $H_{\ell\text{-comp}}^i(X)$ are of finite type. The proof uses the Hochschild–Serre spectral sequence for the covers $X_n \rightarrow X$ and the finiteness theorem for cohomology of proper schemes.

5. Applications

Theorem 5.1.1. Rationality of Z -functions. Let X be of finite type over $\mathbb{F}_q$ and let $\mathcal{F}$ be a constructible $\mathbb{Q}_\ell$ -sheaf on X . Then $Z(X, \mathcal{F}; t)$ is a rational function of t .

When X is proper and smooth this follows from the Lefschetz trace formula 3.2.1. The general case is deduced by dévissage, stratification, and the long exact cohomology sequence.

Theorem 5.2.1. Unit theorem. Let K be a number field, let $\mathcal{O}_K$ be its ring of integers, and let ℓ be a prime. Then $\mathcal{O}_K^\times$ is finitely generated, and its rank is

$$\text{rank}(\mathcal{O}_K^\times) = r_1 + r_2 - 1,$$

where r_1 is the number of real places and r_2 the number of complex places of K .

This classical theorem can be interpreted in terms of ℓ -adic cohomology. The Iwasawa spectral sequence for $\text{Spec}(\mathcal{O}_K[1/\ell])$ gives a description of the unit group in terms of Iwasawa modules.

Theorem 5.3.1. Galois structure of cohomology. Let k be a number field and let X be a smooth projective variety over k . Let $\bar{k}$ be an algebraic closure of k . Then the representation of $\text{Gal}(\bar{k}/k)$ on

$$H_{\text{et}}^i(X \otimes_k \bar{k}, \mathbb{Q}_\ell)$$

is semisimple, and its composition factors are independent of ℓ for ℓ different from the residual characteristics.

The semisimplicity follows from Deligne's theory of weights. The independence of ℓ is a consequence of the Weil conjectures and the theory of algebraic cycles. This completes Exposé VI.